

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a responsive and visually attractive webpage using HTML5 structural elements and CSS3 styling properties.

3. OBJECTIVE

The objectives of this experiment are:

- ☐ ☐ To understand the structure and features of HTML5.
 - ☐ To use semantic HTML5 elements such as <header>, <section>, <aside>, <article> and <footer>.
 - ☐ To create an AI-based health chatbot interface using HTML form elements.
 - ☐ To design an appointment form for collecting patient information.
 - ☐ To apply CSS3 properties for styling the webpage.
 - ☐ To use background gradients, colors, borders, border-radius and box-shadow.
 - ☐ To implement CSS3 transitions and hover effects for interactive elements.
 - ☐ To improve the visual appearance and usability of the webpage.
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4. THEORY

HTML5 is the latest major version of HyperText Markup Language used to structure and present content on web pages. It provides semantic elements that give meaningful structure to a

webpage. Elements such as <header>, <section>, <aside>, <article> and <footer> help organize the webpage into different logical sections.

CSS3 (Cascading Style Sheets Level 3) is used to control the appearance and presentation of HTML elements. CSS3 provides properties for setting colors, backgrounds, fonts, spacing, borders, shadows and other visual effects.

In this experiment, HTML5 is used to create a health-related webpage called **HealthMonAi**. The webpage contains a navigation header, AI chatbot section, About Us section, hospital appointment form and footer.

CSS3 is used to style the webpage using properties such as background-image, linear-gradient(), background-color, padding, box-shadow, border-radius, transition and transform. Hover effects are also applied to buttons and the About Us section to make the webpage interactive.

5. ALGORITHM / PROCEDURE

- ☐ ☐ Create an HTML5 document using the <!DOCTYPE html> declaration.
 - ☐ Create the <head> section and specify the character encoding and viewport settings.
 - ☐ Link the external CSS file to the HTML document.
 - ☐ Create a header section containing the HealthMonAi title and navigation links.
 - ☐ Create an AI Chatbot section with a heading, description, text area and Enter button.
 - ☐ Create an About Us section using the <aside> element.
 - ☐ Add an ordered list and health-related image to the About Us section.
 - ☐ Create the Hospital appointment section using the <article> element.
 - ☐ Create an appointment form using <form> and <fieldset>.
 - ☐ Add input fields for patient name and phone number.
 - ☐ Add radio buttons for selecting gender.
 - ☐ Add a drop-down list for selecting the location.
 - ☐ Add a textarea for entering the patient's problem.
 - ☐ Add a Submit button to the appointment form.
 - ☐ Create a footer containing a thank-you message.
 - ☐ Apply CSS3 styling to the webpage using colors, gradients, shadows, rounded corners and spacing.
 - ☐ Apply hover effects and transitions to interactive elements.
 - ☐ Save both the HTML and CSS files and open the HTML file in a web browser.
 - ☐ Verify that all sections, form controls and CSS effects are displayed correctly.
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6. PROGRAM

Design.html

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML&CSS</title>
    <link rel="stylesheet" href="css2.css" target="">
</head>


<body>
    <header>
        <table class="header_class" width="100%">
            <tr>
                <td width="100%">
                    <center>
                        <h1>HealthMonAi</h1>
                        <nav class="navg">
                            <a href="#Home">Home</a>&nbsp;&nbsp;&nbsp;;|&nbsp;&nbsp;&nbsp;
                            <a href="#Chatbot">Chatbot</a>&nbsp;&nbsp;&nbsp;;|&nbsp;&nbsp;&nbsp;
                            <a href="#Application">Appointment</a>&nbsp;&nbsp;&nbsp;;|&nbsp;&nbsp;&nbsp;
                            <a href="#about">>About Us</a>&nbsp;&nbsp;&nbsp;;
                        </nav>
                    </center>
                </td>
            </tr>
        </table>
    </header>
    <section class="sec_class" id="Chatbot">
        <center>
            <h1>AI Chatbot</h1>
            <p>This chatbot helps you to clarify your doubts related to health</p>
            <p><label for="chat"></label>
                <textarea name="chatbot" id="chat" cols="70" rows="5"
                    placeholder="ask your question here..." style="background-color: darkgrey; border-radius: 10px;"></textarea><br>
                <input type="submit" value="Enter" class="button_class" >
            </p>
        </center>
    </section>
    <hr>
    <table width="100%" border="0" cellspacing="10" cellpadding="0">
        <tr>
            <td width="30%">
                <aside class="aside class" id="about">
```

```

<h1>About Us</h1>
<ol>
  <li>It is most usefull app used now a days</li>
  <li>Health is the most preffered case by all</li>
  <li>Now we have integrated healh with Ai</li>
  <li>We do have a specific chatbot for it</li>
</ol>

</aside>
</td>
<td width="70%">
  <article class="article_class">
    <center>
      <h1>Hospital</h1>
    </center>
    <form class="form_class" id="Application" action="#" method="">
      <fieldset>
        <legend>Appointment Form</legend>
        <p>
          <label for="name">Name:</label>
          <input type="text" id="name" name="patient name" required
            placeholder="enter patients name..">
        </p>
        <p>
          <label for="phn">Phone:</label>
          <input type="number" name="phn number" id="phn" required>
        </p>
        <p>
          <label for="gender">Gender:</label><br>
          <input type="radio" id="male" name="gender" value="male">
          <label for="male">Male</label><br>

          <input type="radio" id="female" name="gender" value="female">
          <label for="female">Female</label><br>

          <input type="radio" id="other" name="gender" value="other">
          <label for="other">Other</label>

        </p>
        <p>
          <label for="location">Location:</label>
          <select name="adress" id="location">
            <option value="Vizag">Vizag</option>

```

```

        <option value="Hyd">Hyd</option>
        <option value="bangl">Bannglore</option>
        <option value="chenn">Chennai</option>
    </select>
</p>
<p>
    <label for="textbox">Problem:</label>
    <textarea name="textb" id="textbox" placeholder="type your problem
here...."></textarea>
</p>
    <button type="submit" class="button_class">Submit</button>
</fieldset>
</form>
</td>
</article>
</tr>
</table>
<hr>
<footer class="footer_class">
    <center>
        <h1>Thank you for using this</h1>
    </footer>
</center>
</body>

</html>

```

Style.css

```

body{
    font-family: 'Times New Roman', Times, serif;
    background-image: linear-gradient( to right,rgb(129, 134, 132) 30%, rgb(142, 189, 161) 70%);
/*background color and to used to flow of colors*/
}
.header_class{
    background-color: mediumseagreen;
    padding: 15px;
    box-shadow: 10px 5px rgba(148, 183, 146, 0.5);
    border-radius: 20px;
}
.aside_class{
    background-color: mediumseagreen;
    border-radius: 20px;
    transition: all 0.3s ease; /*smoooth transition for 0.3secs*/
}

```

```

.aside_class:hover{
    background-color: darkseagreen;
    cursor: pointer;
    transform: scale(1.02); /*how much to blow out*/
}
.button_class{
    background-color: crimson;
    border-radius: 10px;
    padding: 6px 10px;
    color: aliceblue;
    cursor: pointer;
    transition: all 0.3s ease;
}
.button_class:hover{
    background-color: lightcoral;
    transform: scale(1.1);
}

```

7. CODE EXPLANATION

HTML Code:

- **<!DOCTYPE html>** – Declares the document as an HTML5 document.
- **<head>** – Contains metadata, viewport settings and the webpage title.
- **<link rel="stylesheet">** – Connects the external CSS file with the HTML webpage.
- **<header>** – Defines the header section containing the website name and navigation menu.
- **<nav>** – Contains links to different sections of the webpage.
- **<section>** – Defines the AI Chatbot section.
- **<textarea>** – Provides an area for the user to enter health-related questions.
- **<input type="submit">** – Creates the Enter button for the chatbot.
- **<aside>** – Defines the About Us sidebar containing information and an image.
- **<article>** – Defines the hospital appointment content.
- **<form>** – Creates the appointment form for collecting patient details.
- **<fieldset>** – Groups the appointment form controls together.
- **<legend>** – Provides a title for the appointment form.
- **<input type="text">** – Accepts the patient's name.
- **<input type="number">** – Accepts the patient's phone number.
- **<input type="radio">** – Allows the user to select gender.
- **<select>** – Provides a drop-down list for selecting location.
- **<textarea>** – Allows the user to enter details about the problem.

- **<button>** – Creates the Submit button.
- **<footer>** – Defines the footer section of the webpage.

CSS Code:

- **font-family** – Sets the font style of the webpage.
- **background-image** – Applies a background image to the webpage.
- **linear-gradient()** – Creates a smooth gradient background using multiple colors.
- **background-color** – Sets the background color of different elements.
- **padding** – Adds space inside an element.
- **box-shadow** – Adds a shadow effect around an element.
- **border-radius** – Creates rounded corners.
- **transition** – Produces a smooth transition when CSS properties change.
- **:hover** – Applies styling when the mouse pointer is placed over an element.
- **transform: scale()** – Enlarges an element when the hover effect is activated.
- **cursor: pointer** – Changes the mouse cursor to a pointer when hovering over interactive elements.

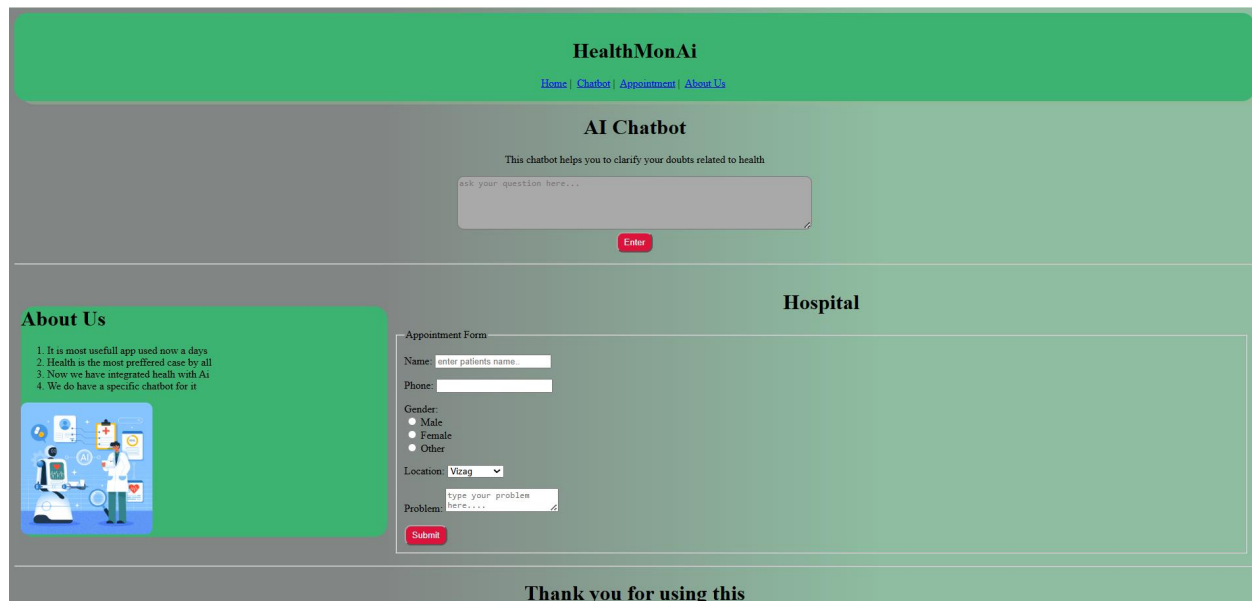
8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	HealthMonAi webpage should be displayed	Webpage displayed correctly	Pass
TC-02	Click the Home navigation link	Home section should be accessed	Home section accessed	Pass
TC-03	Enter a health-related question in the chatbot text area	Question should be entered successfully	Question entered successfully	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-04	Enter patient name and phone number	Input fields should accept the data	Data accepted successfully	Pass
TC-05	Select a gender option	Selected gender should be displayed	Gender selected successfully	Pass
TC-06	Select a location from the drop-down list	Selected location should be displayed	Location selected successfully	Pass
TC-07	Enter a problem in the textarea	Problem description should be accepted	Description accepted successfully	Pass
TC-08	Hover over the About Us section	Section should change appearance with a smooth effect	Hover effect displayed	Pass
TC-09	Hover over the Submit/Enter button	Button should change appearance and scale smoothly	Hover effect displayed	Pass
TC-10	Click the Submit button after entering valid details	Form should be submitted	Form submitted successfully	Pass

9. OUTPUT



10. RESULT

Thus, a visually attractive webpage was successfully designed using **HTML5** and **CSS3**. Semantic HTML5 elements were used to structure the webpage, while CSS3 properties such as gradients, colors, shadows, rounded corners, transitions and hover effects were used to enhance its appearance and interactivity. The chatbot interface and hospital appointment form were also successfully implemented.